

Name _____ Date _____ Period _____

Domain and Range: All Three Notations

Algebra II Honors | Unit 1 | Day 1 | TEK A2.7.I

Objective: I can write domain and range in verbal, inequality, set, and interval notation, determine a domain algebraically from a function rule, and use union notation for a domain that comes in separate pieces.

Vocabulary

Set-builder notation — $\{x \mid \text{condition}\}$, read "the set of all x _____ the condition holds."

Interval notation — _____ includes an endpoint, _____ excludes it. Infinity always takes a _____.

Union $\cup$ — joins two separate intervals when the domain has a _____ in it.

Undefined — an input the rule cannot accept. Two causes in this course: a _____ and a _____.

Part 1 — From Pairs and Tables

List least to greatest, each value once.

a. $\{(-4, -6), (0, -1), (2, 4), (3, -7), (7, 7)\}$ D _____ R _____

b. $\{(-5, 3), (-1, 0), (2, 3), (6, -8)\}$ D _____ R _____

c. Why does the range in (b) have three values when there are four pairs? _____

Part 2 — All Three Notations

Complete every cell.

Verbal description	Inequality	Set notation	Interval notation
all real numbers	_____	_____	_____
all reals at least -6	_____	_____	_____
all reals less than 5	_____	_____	_____
all reals from -3 to 3, 3 included only	_____	_____	_____
all reals except 2	_____	_____	_____
just the numbers -2, 0, and 4	(none)	_____	_____

The last two rows are the ones to think hardest about. Explain the difference between them:

Row 5 is continuous with one point removed; row 6 is _____

Part 3 — Finding a Domain from the Rule

No graph, no table - reason from the algebra. Ask what the rule forbids, then exclude it.

Function	What is forbidden?	Inequality	Interval notation
$f(x) = \sqrt{x - 3}$	_____	_____	_____
$g(x) = 1/(x - 5)$	_____	_____	_____
$h(x) = \sqrt{2x + 8}$	_____	_____	_____
$k(x) = 1/\sqrt{x - 1}$	_____	_____	_____

Look at $k(x)$. Why is the answer $x > 1$ and not $x \geq 1$?

At $x = 1$ the radicand is 0, and that would make the denominator _____ — so 1 must be excluded too.

Part 4 — A Domain in Three Pieces

Find the domain of $m(x) = (x + 2)/(x^2 - 9)$. Show the factoring.

Denominator factors as _____, so x cannot be _____

Domain, interval notation: _____

Domain, set notation: _____

Which notation communicates this domain more efficiently, and why?

Part 5 — Going Further

1. Explain why ∞ can never take a bracket, no matter the function.

2. Write a function whose domain is exactly $[4, \infty)$. _____

3. Write a function whose domain is exactly $(-\infty, 0) \cup (0, \infty)$. _____

4. Can a domain be empty? Give a rule with no valid inputs, or argue that none exists.

Sample: _____

Exit Ticket

Give the domain of $f(x) = \sqrt{x + 5}/(x - 2)$ in interval notation, and justify each restriction.

Domain _____

Two reasons: _____